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Laxative and purgative actions of phytoactive compounds from beetroot juice against loperamide-induced constipation, oxidative stress, and inflammation in rats

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Abstract

Background: Chronic constipation is a gastrointestinal functional disorder which affects patient quality of life. Therefore, many studies were oriented to search herbal laxative agents. In this study, we investigated the phytochemical composition of beetroot juice (BJ) and its laxative potential in an experimental model of constipation and colonic dysmotility induced by loperamide (LOP) in *Wistar* rats.

Methods: Animals were concurrently pretreated with LOP (3 mg/kg, *b.w.*, *i.p.*) and BJ (5 and 10 mL/kg, *b.w.*, *p.o.*), or yohimbine (2 mg/kg, *b.w.*, *i.p.*), during 1 week. The laxative activity was determined based on the weight, frequency, and water content of the feces matter. The gastric-emptying test and intestinal transit were determined. Colon histology was examined, and oxidative status was evaluated using biochemical-colorimetric methods.

Key results: The in vivo study revealed that LOP induced a significant inhibition of gastrointestinal motility, negative consequences on defecation parameters, oxidative stress, and colonic mucosa lesions. Conversely, administration of BJ reestablished these parameters and restored colonic oxidative balance. Importantly, BJ treatment protected against LOP-induced inflammatory markers (pro-inflammatory cytokines and WBC) and the increase in intracellular mediators such as hydrogen peroxide, free iron, and calcium levels.

Conclusions & inferences: This study demonstrated that the bioactive compounds in BJ provided an anti-constipation effect by modulating intestinal motility and regulating oxidative stress and inflammation induced by LOP intoxication.

KEYWORDS
beetroot juice, *Beta vulgaris*, constipation, gastrointestinal motility, inflammatory markers, oxidative stress, rats

1 | INTRODUCTION

Chronic constipation, a typical gastrointestinal functional disorder affecting 3%–15% of the general population, leads to discomfort and a diminished quality of life.[1] This intestinal disorder, impacts individuals of all ages, with prevalence rates ranging from 0.7% to 29.6% in children and adolescents to 15%–50% in the elderly.[2,3] It is characterized by infrequent or difficult defecation (occurring once every 3–4 days or less), incomplete fecal excretion, or stool hardness, constipation poses risks of colonic cancer, hemorrhoids, and other gastrointestinal complications such as nausea, vomiting, restlessness, stomach obstruction, perforation, as well as aspiration or fatal pulmonary embolism.[4,5] Contributing factors include insufficient fiber and fluid intake and adverse drug reactions.[6] Epithelial goblet cells play a role in maintaining the intestinal mucosal barrier by secreting a hydrating gel, protecting against mechanical and chemical irritants. Direct damage to epithelial cells can compromise this barrier function.[7]

Loperamide is extensively employed in clinical settings to manage acute and chronic diarrhea owing to its well-documented efficacy in modulating intestinal motility.[8] LOP-induced constipation, a model frequently utilized in animal experiments,[9,10] arises from delayed colonic transit and is identified as spastic constipation in humans due to reduced stool frequency and heightened colonic contractions. Loperamide reduces intestinal water secretion and colonic peristalsis, which prolongs the time for fecal evacuation and slows down the movement of intestinal contents.[9]

Medications containing sennoside or magnesium oxides, such as senna, correctol, exlax, and senokot, are commonly used for constipation and exhibit potent laxative/purgative effects.[11] These drugs typically regulate gastrointestinal tract motility and facilitate stool passage. However, the use of these drugs is constrained by high costs and side effects like severe diarrhea, with frequent use potentially leading to melanosis coli, a colorectal neoplasm risk factor.[8] Opioid-induced constipation (OIC) affects approximately 41%–81% of patients receiving opioids for non-cancer pain. OIC can have a significant impact on patients' quality of life, particularly in terms of functionality and well-being.[12] Consequently, treating and preventing constipation remains a current challenge, with research focused on identifying drugs with similar efficacy but fewer side effects. Natural products are gaining attention in this regard, as consumers increasingly favor “functional foods” for their health benefits and antioxidant properties.[13,14] Numerous studies have reported that certain plant extracts effectively enhance gastrointestinal motility.[9,10]

Beta vulgaris var. rubra L., commonly referred to as red beetroot, is a vegetable that belongs to the Chenopodiaceae family. It is widely distributed around the world and is traditionally used for producing red juice and natural pigment on a commercial scale. Primarily grown for its roots, which serve as both a food source and a natural dye, beetroot has evolved into a functional food.[15,16] Recognized for its nutritional value, it is rich in dietary fiber, nitrates, minerals such as potassium, sodium, iron, copper, magnesium, calcium, phosphorus, and zinc, as well as vitamins like retinol and B-complex, antioxidants, and betalain pigments. These pigments, which are divided into red betacyanins and yellow betaxanthin, are responsible for the vibrant colors of beetroot.[17] Previous research has highlighted the potential antioxidant properties of betalains and their absorption in humans.[18,19] Several studies have also investigated beetroot's effects on various health conditions, including type 2 diabetes, cardiovascular diseases, hypertension, dyslipidemia, and inflammation.[20–22]

Hence, the aim of this study was to evaluate the purgative/laxative effects of beetroot juice (BJ) on gastrointestinal-physiological functions against LOP-induced constipation, oxidative stress, and inflammation in rats.

Key points

- Extracts of red beetroot juice (BJ) have enhanced a laxative product used to treat constipation in rats. This research supports the use of this plant as a natural remedy instead of chemical drugs.
- Constipation induced by loperamide in 50 rats was followed by histological and biochemical tests to evaluate the effects on oxidative stress and inflammation.
- BJ showed an acceleration of gastric emptying and gastrointestinal transit in vivo. Moreover, BJ modulated oxidative stress through antioxidant enzyme regulation and decreased colon inflammation by modulating pro-inflammatory cytokines in the colonic mucosa.

2 | MATERIALS AND METHODS

2.1 | Reagents

Red phenol, hematoxylin, eosin, formaldehyde, 5.5-dithiobis (2-nitrobenzoic acid) (DTNB), trichloroacetic acid (TCA), butylhydroxytoluene (BHT), methanol, epinephrine, NaCl, NaOH, charcoal meal, gum arabic, and bovine catalase were from Sigma Chemical Co. (Sigma-Aldrich GmbH, Steinheim, Germany). Sodium bicarbonate ($NaHCO_3$) and hydrogen peroxide (H_2O_2) were purchased from Biomaghreb Society (Ariana, Tunisia) and were ISO 9001 certified. Loperamide and Dulcolax were purchased from Tunisia's central pharmacy.

2.2 | Plant collection and beetroot juice (BJ) preparation

The beetroots were collected from the mountains area of BouSalem (Jendouba Governorate, North West of Tunisia) in March and April 2022. A voucher sample was registered in the herbarium of the Higher Institute of Biotechnology in Beja, Jendouba University, assigned the identification number BR-ISBB-10/05/22.

After collection, the beetroots were meticulously cleansed with tap water and then homogenized using a household mixer equipped with an integrated centrifuge (Braun multi-press). The resulting juice underwent centrifugation at 10000 g for 15 min. Subsequently, the supernatants were filtered through a 0.22 μm cellulose acetate filter. The resulting juice had 90.35 ± 0.78% moisture content without any chemical additives. The obtained juice was immediately stored at −80°C in the dark until further use.

2.3 | Biochemical and phytochemical analyses

2.3.1 | Mineral determination

The BJ mineral contents were assessed using the procedure outlined by Jian et al.[23] A total of 27.1 mL of BJ solution underwent digestion in 8 mL of nitric acid (HNO_3) for 16 h at room temperature. Subsequently, the sample was subjected to the block host (ED54 Digiblock Digester Lab Tech, USA) at 133°C until acid volatilization. Following the cooling phase, 2 mL of distilled water was introduced, and the sample was returned to the block at the same temperature. Subsequently, 40 mL of distilled water was incorporated and filtered using a syringe filter (0.4 μm pore diameter). The concentrations of minerals in the samples were determined through atomic absorption spectrophotometer analysis (AVANTA GBC Scientific Equipment Pty. Ltd., Australia).

2.3.2 | Total polyphenol content (TPc)

Total phenolic content was determined following the method described by Haseeb et al.[24] Indeed, 50 μL of the extract was combined with 120 μL of Folin–Ciocalteu reagent and 2 mL of distilled water. After 5 min, 375 μL of 10% (w/v) sodium carbonate solution was introduced. The reaction mixture was then kept in the dark for 2 h at room temperature, and the absorbance was measured at 765 nm using a UV–VIS spectrophotometer. Gallic acid served as the phenolic standard, and the results were expressed as mg of gallic acid equivalents (GAE)/g fresh matter (FM).

2.3.3 | Total flavonoid contents (TFc)

Total flavonoid concentration was determined using the aluminum chloride ($AlCl_3$) colorimetric method.[25] In brief, 400 μL of the sample was mixed with 120 mL of $NaNO_2$ (5%, w/v), and then 120 mL of $AlCl_3$ (10%, w/v) was added. After 6 min, 800 μL of NaOH (1 M) was introduced. The absorbance was measured at 510 nm using a UV–VIS spectrophotometer. Catechin served as the condensed flavonoid standard for quantifying the total flavonoid content, and the results were expressed as mg quercetin equivalents (QE)/LFM (Liquor Fresh Matter).

2.3.4 | Total tannin content (TTc)

Total Tannin content was determined following the method outlined by Kujala et al.[26] In summary, 0.5 mL of Folin–Ciocalteu (50%) was added to 0.5 mL of BJ and incubated for 5 minutes in the dark. Subsequently, the mixture was combined with 1 mL of 20% sodium carbonate solution (50%), incubated for 10 min at room temperature, and then centrifuged at 1000 g for 5 min. The resulting mixture was measured at 730 nm using a UV–VIS spectrophotometer. Tannic acid was employed as the condensed tannin standard for quantifying the total tannin content. The TTC of BJ was expressed as mg tannic acid equivalent (TAE) per liter of fresh matter (mg TAE/L FM).

2.3.5 | Condensed tannin content (CTc)

CTc in BJ followed a modified vanillin assay.[27] This assay involved the reaction between vanillin and condensed tannin molecules in an acidic environment, leading to the formation of a complex read at 500 nm. Condensed tannin content was determined in mg of catechin equivalent (CE) per liter of fresh matter (mg CE/L FM).

2.3.6 | Total carbohydates (TC) and fiber contents

TC was determined using the dinitrosalicylic acid (DNS) reagent.[28] Briefly, 400 μL of DNS reagent was added to 100 μL of BJ sample and incubated at 95°C for 5 min. The absorbance was measured at 540 nm using a UV–VIS spectrophotometer. The test was carried out in triplicate. The fiber content was determined by AOAC Method 991 (1995).

2.4 | Antioxidant activity

2.4.1 | 2-Diphenyl-1-picrylhydrazyl (DPPH$^{\bullet}$) radical scavenging capacity

The scavenging capacity of BJ for DPPH was evaluated according to Bersuder et al.[29] Briefly, 500 μL of the test sample was mixed with 375 μL of ethanol (100%) and 125 μL of DPPH (0.02 Mm in ethanol). The mixture was vortexed, then left for 60 min in the dark at room temperature. The control was conducted in the same manner, except that distilled water was used instead of the sample. Ascorbic acid was used as a standard.

The absorbance was read at 517 nm, and the percentage of DPPH radical scavenging activity was calculated using the following formula:

$$\mathrm{RSA}\,(\%) = \frac{(\mathrm{AC} - \mathrm{AS})}{\mathrm{AC}} \times 100$$

with, AC and AS represent, respectively, the absorbance of the control and the sample reaction. The results of the RSA were shown as a half-maximal inhibitory concentration (IC_{50}) value (μg/mL).

2.4.2 | Ferrous ion-chelating ability assay

The assessment of the Fe^{2+} chelation activity of BJ was conducted utilizing the method described by Decker and Welch[30] with a slight modification. In brief, 50 μL of $FeCl_2$ $4H_2O$ solution (2 mM) was introduced to 50 µL of BJ, which was then diluted in 450 µL of distilled water. The reaction commenced with the addition of 5 mM ferrozine (2 mL), and the resulting solution was vigorously shaken before incubating for 10 min at room temperature. The absorbance was measured at 562 nm, and the Fe^{2+} chelation capacity was subsequently calculated relative to the control sample. The percentage of Fe^{2+} chelation activity (%) was determined using the following formula:

$$\% \, Fe^{2+} \text{ Chelation} = ((Ac + Ab - As) / Ac) \times 100$$

with, Ac represents the absorbance of the control reaction, Ab is the absorbance of the blank, and As is the absorbance of the sample.

2.5 | Induction of constipation

2.5.1 | Animals

Physiological studies were performed on male *Wistar* rats (weighing 220 ± 20 g, 15 weeks old, housed 5 per cage) and male *Swiss Albino* mice (weighing 25 ± 5 g, housed 8 per cage) were procured from the Society of Pharmaceutical Industries of Tunisia (SIPHAT, Tunis, Tunisia). The animals were accommodated in communal cages under standard conditions (12/12 h light–dark cycle, temperature maintained at 22 ± 2°C) and were provided with standard rodent feed (BADR, Utique, TN) and water ad libitum. Additionally, all animal procedures were ethically approved by the Institutional Animal Care and Use Committee of the National Institute of Health and were conducted in accordance with the NIH Guidelines for the Care and Use of Laboratory Animals.[31]

2.5.2 | Acute toxicity of BJ in animal mice model

The acute toxicity of BJ (presumably referring to the substance being tested) was evaluated through oral administration to male *Swiss Albino* mice. A series of increasing doses of BJ (0.25, 0.5, 1, 2.5, 5, 10, 25, and 50 mL/kg b.w.) were prepared and orally administered to 8 groups of male mice ($n = 10$ in each). A control group was treated with 10 mL/kg of NaCl 0.9%. Observations, encompassing mortality, mobility, respiratory changes, and appetite, were conducted at intervals, with examinations every 30 min during the initial 6 h on the first day, followed by once-daily assessments for 48 h.

2.5.3 | Constipation model

A total of 50 *Wistar* rats were divided into 5 groups ($n = 10$), including the normal control, constipated rats, the constipated-BJ-treated group (5 and 10 mL/kg, b.w.). It was indicated that doses of 5 and 10 mL/kg body weight of BJ were the lowest doses that provided significant protective effects without side effects. Finally, the positive control group was treated with yohimbine (2 mg/kg). All animals, except the normal control group, were orally administered 3 mg/kg loperamide hydrochloride dissolved in saline solution, once daily for 7 continuous days, 1 h before administration of the test substances to induce constipation as described previously. The normal control rats were administered a 0.9% saline solution (2 mL/kg), the other treatment groups (those treated with BJ and YOH) received the specified dose once daily. At the end of the experiment; all animals were anesthetized with ketamine (70 mg/kg, b.w.i.p.) and sacrificed. The distal colon of each animal was removed, cut longitudinally, cleaned with physiological saline solution, and homogenized on ice with phosphate-buffer saline (PBS, pH 7.4). Samples were then centrifuged at 9000 *g* for 15 min at 4°C, and supernatants were collected. Blood samples with the addition of EDTA were collected and centrifuged at 3000 *g* for 20 min at 4°C to obtain plasma. Plasma and supernatant samples were stored at −80°C for further biochemical analysis.

2.5.4 | Body weight, food intake, water consumption, and fecal parameter measurements

Throughout the experiment and daily, the weight of the rats, as well as their water and food intake, were documented each day at 9:00 a.m. This was carried out consistently using an electric balance and a measuring cylinder for the entire duration of the experiment. To ensure precision, all measurements were conducted three times.

Similarly, 1 h after the administration of the extract and drug each day, the fecal pellets expelled were directly collected. The wet weight and water content of these collected pellets were then analyzed. The water content was determined by subjecting the fecal pellets to a drying process at 70°C for 24 h in an oven, followed by weighing.[32] The water content of the pellets was calculated using the formula:

$$\text{Fecal water content}(\%) = \frac{\text{fecal wet weight} - \text{fecal dry weight}}{\text{fecal wet weight}} \times 100$$

Moreover, the excreted fecal pellets over 24 h were also gathered, and their properties were determined.

2.5.5 | Gastrointestinal transit

The charcoal meal test was used to assess gastrointestinal transit (GIT).[33] A total of 50 rats were divided into five groups, each consisting of 10 rats. The animals underwent a 16-h fasting period with free access to water. The negative control group (Group 1) was given a saline solution (0.9% NaCl, 1 mL/kg, p.o.). Group 2 received loperamide (3 mg/kg, b.w.), while Groups 3 and 4 were pretreated with BJ at varying doses (5, 10 mL/kg, b.w.). Group 5 was administered the standard drug Yohimbine (YOH, 2 mg/kg, b.w.).

After a 2-h interval, all rats were given a charcoal meal (10% charcoal in 5% gum Arabic). Thirty minutes later, the rats were anesthetized and sacrificed, and the small intestine was carefully removed from the abdominal cavity. The distance traveled by the charcoal meal from the pylorus was measured to calculate GIT using the formula:

$$\mathrm{GIT}\,(\%) = \left(\frac{\text{Distance traveled by the charcoal meal (cm)}}{\text{Total length of the small intestine (cm)}}\right) \times 100$$

2.5.6 | Gastric emptying

The gastric emptying rate was assessed using the red phenol method.[34] Thirty animals were distributed into 5 groups, each comprising 10 animals. Group 1, serving as the negative control, received 1 mL of physiological solution (NaCl, 0.9%, p.o.). Group 2, the positive control, was pretreated with LOP (3 mg/kg, b.w.). Groups 3 and 4 underwent pretreatment with BJ at varying doses (5 and 10 mL/kg, b.w.), while Group 5 received the standard drug YOH (2 mg/kg, b.w.). After a 1-h interval, a test meal (1.5 mL/rat) consisting of 50 mg phenol red in 100 mL aqueous methyl cellulose (1.5%) was orally administered.

Phenol red recovered from another group of animals sacrificed immediately after the administration of the test meal served as the standard (0% emptying), while the remaining animals were sacrificed 1 h later. The stomach and its contents were homogenized with 100 mL of NaOH (0.1 N). After allowing the suspension to settle for 1 h at room temperature, 5 mL of the supernatant was combined with 0.5 mL of 20% (w/v) trichloroacetic acid (TCA) and then centrifuged at 1800 g for 20 min. Subsequently, 4 mL of NaOH (0.5 N) was added to the supernatant, and the absorbance of the sample was measured at 560 nm. The percentage of gastric emptying was calculated using the formula:

$$\text{Gastric emptying rate}\,(\%) = \frac{1 - \text{absorbance of treated}}{\text{absorbance of standard}} \times 100$$

2.5.7 | Measurement of serum biochemical parameters

The potential protective effect of BJ against LOP-induced constipation was assessed by examining the plasma concentrations of glucose, total cholesterol (TC), high-density lipoprotein cholesterol (HDL), and triglyceride (TG). Additionally, electrolyte levels, including sodium, potassium, and magnesium, were determined. All analyses were conducted using commercially available diagnostic kits purchased from Biomaghreb, TN.

2.6 | Measurement of oxidative stress biomarkers

The oxidative stress status of the colonic mucosa was assessed by the estimation of malondialdehyde (MDA), hydroxyl radical ($OH^{\bullet}$), thiol groups (-SH), reduced glutathione (GSH), and protein levels, as well as the measurement of antioxidant enzyme activities such as superoxide dismutase (SOD), catalase (CAT), and glutathione peroxidase (GPx).

2.6.1 | Lipid peroxidation determination

Lipid peroxidation was assessed by employing the method devised by Draper and Hadley.[35] This method evaluates damage to polyunsaturated fatty acids in lipids. In brief, tissue homogenates of colonic mucosa were combined with a BHT-TCA solution (1% BHT w/v dissolved in 20% TCA w/v) and then subjected to centrifugation at 1000 g for 5 min at 4°C. The resulting supernatants were mixed with 0.5 N HCL and 120 mM TBA in 26 mM Tris, followed by heating in a water bath at 80°C for 10 min.

The absorbance was determined at 532 nm, and the levels of malondialdehyde (MDA) were expressed in nmol/mg protein, utilizing a molar extinction coefficient of $1.56 \times 10^5\,M^{-1}\,cm^{-1}$.

2.6.2 | Superoxide dismutase (SOD) activity

The superoxide dismutase (SOD) activity was determined following the procedure outlined in Misra and Fridovich's method.[36] This technique is based on the inhibition of the auto-oxidation of epinephrine to adenochrome at pH 10.2. In brief, the assay mixture comprised 10 μL of bovine catalase (0.4 U/mL), sodium carbonate/bicarbonate buffer (62.5 mM; pH 10.2), and 20 μL of epinephrine (5 mg/mL).

One unit (U) of SOD is defined as the enzyme necessary to inhibit the rate of epinephrine autoxidation by 50%. The enzymatic activity of SOD was expressed as U/mg protein.

2.6.3 | Catalase (CAT) activity

Catalase (CAT) activity was evaluated by monitoring the reduction of H_2O_2 levels at 240 nm.[37] In short, the reaction mixture consisted of 33 mM H_2O_2, 50 mM phosphate buffer (pH 7), and 30 μL of tissue extract. CAT activity was quantified as μL H_2O_2 consumed per minute per milligram of protein, employing the extinction coefficient of H_2O_2 ($40\,mM^{-1}\,cm^{-1}$).

2.6.4 | Glutathione peroxidase (GPx) activity

The glutathione peroxidase (GPx) activity was determined following the method outlined by Flohé and Günzler.[38] The reaction mixture consisted of 0.2 mL of phosphate buffer (0.1 M, pH = 7.4), 0.2 mL of mM reduced glutathione (GSH), 0.4 mL of H_2O_2 (5 mM), and 0.2 mL of tissue homogenate. The mixture was incubated for 1 minute at 37°C, and the reaction was halted by adding 0.5 mL of 5% TCA.

After centrifugation at 1500 g for 5 min, 0.5 mL of the supernatant was combined with 0.5 mL of phosphate buffer and 0.5 mL of DTNB (10 mM). The absorbance was measured at 412 nm. GPx activity was expressed as nmol of GSH consumed per minute per milligram of protein, utilizing the extinction coefficient of $1.36 \times 10^3\,M^{-1}$.

2.7 | Non-enzymatic antioxidant levels assessment

2.7.1 | Total thiols group (-SH) contents

The determination of total thiol group (-SH) contents was carried out using Ellman's method.[39] In brief, aliquots of colonic mucosa were combined with 100 μL of SDS (10%) and 800 μL of phosphate buffer (10 mM, pH 8.2), and the initial optical density was measured at 412 nm (A_1). Subsequently, 100 μL of 10 mM 5,5′-dithiobis-(2-nitrobenzoic acid) (DTNB) was added, and the mixture was incubated at 37°C for 15 min. After incubation, the absorbance of the sample was measured again at 412 nm (A_2).

The thiol group content was calculated from A1 to A2 and expressed as nmol of thiol groups per milligram of protein, utilizing a molar extinction coefficient of $13.6 \times 10^3\, M^{-1}\, cm^{-1}$.

2.7.2 | Reduced glutathione concentrations

The concentrations of reduced glutathione were determined using the method described by Sedlak and Lindsay.[40] In brief, 5 mL of supernatant was mixed with 4 mL of cold distilled water and 1 mL of trichloroacetic acid (TCA, 50%). The mixture was then vortexed for 10 min and centrifuged at 1200 ***g*** for 15 min. Subsequently, 2 mL of the recovered supernatant was added successively to 4 mL of Tris–HCl buffer (0.4 M, pH 8.9) and 100 μL of 5,5′-dithiobis-(2-nitrobenzoic acid) (DTNB, 0.01 M). The absorbance was rapidly measured at 412 nm against the blank, which contained only the buffer.

2.8 | Intracellular mediator determinations

2.8.1 | Peroxide hydrogen determination

The quantification of H_2O_2 concentrations in colon mucosa was carried out using Dingeon's method.[41] In this method, in the presence of phenol, hydrogen peroxide, and peroxidase, H_2O_2 reacts with p-hydroxybenzoic acid and 4-aminoantipyrine, forming a complex known as quinoneimine. This complex exhibits a pink color, which can be detected spectrophotometrically at 505 nm.

2.8.2 | Calcium and free iron evaluation

Calcium (Ca^{2+}) contents in colonic mucosa were determined using the method described by Stern and Lewis.[42] In this method, calcium (Ca^{2+}), which precipitates as Ca^{2+} oxalate, forms a complex with o-cresol phthalein, and the reaction is colorimetrically measured at 570 nm.

Iron contents were determined using a ferrozine reagent, following the method described by Leardi et al.[43] Briefly, ferrozine combines with iron, which has been produced from the transferrin-iron complex by guanidine acetate solution and reduced by ascorbic acid, to create a pink complex. This complex can be detected at 562 nm.

2.9 | Measurement of inflammation parameters

2.9.1 | Pro-inflammatory cytokines

TNF-α and IL-1β levels were quantified in colonic mucosa supernatants using enzyme-linked immunosorbent assay (ELISA) rat kits.[44] Briefly, anti-rat's monoclonal antibodies were coated into 96-well plates and incubated overnight at 4°C. After additional washings, samples or IL-Iβ and TNF-α standard were added and incubated for 2 h at room temperature. After washing the wells, biotinylated anti-rat's IL-Iβ and TNF-α antibodies were added, followed by 2 h of incubation. Then, 100 μL of hydrogen peroxide and tetramethyl benzidine were added, initiating the color reaction. The reaction was stopped by adding phosphoric acid (1 M). The absorbance was measured at 450 nm using an ELISA microplate reader. The results were expressed as pg. per mg of proteins. The results were standardized by comparison with a standard curve. The assays were performed in triplicate.

2.9.2 | Total differential leukocytes measurement

To assess the toxic stress-induced, the white blood cells (WBC), lymphocytes (LYM), monocytes (MON), neutrophils (Neu), and eosinophils (EOS) were analyzed. All tests were performed using a fully electronic automated clinical biochemistry analyzer (HORIBA-ABX Pentra XL 80).

2.10 | Histopathological study

Following sacrifice, colonic tissue fragments from each group were collected and fixed in 10% paraformaldehyde. Subsequently, they underwent dehydration in a graded series of ethanol. The tissues were then embedded in a paraffin block, sectioned at a thickness of 5 μm, deparaffinized, and rehydrated. Staining was performed using hematoxylin and eosin (H&E) according to standard histological procedures.[10] The histopathological changes in the colon were observed under a microscope (Leica DM4000B), and images were recorded with a camera (Leica DFC300 FX) for further analysis.

2.11 | Statistical analysis

All experimental data were analyzed using the professional edition of the GraphPad Prism 8.0.2 software package (San Diego, CA, USA) and expressed as means ± standard error of the mean (SEM). The data represent observations from six to ten distinct instances. Statistical differences between the groups were assessed with a one-way analysis of variance (ANOVA), followed by the Tukey post hoc test for multiple comparisons. Differences were considered statistically significant at $p < 0.05$.

3 | RESULTS

3.1 | Phytochemical evaluation

The outcomes of the quantitative assessment of secondary metabolite content are detailed in Table 1. The beverage (BJ) is rich in various secondary metabolites, including elevated levels of polyphenols, flavonoids, total tannins, condensed tannins, and total carbohydrates. Additionally, BJ serves as an outstanding source of dietary fibers, measuring 14.74 ± 1.25 g/L. Moreover, the composition of mineral elements (Table 1) reveals that BJ exhibits higher concentrations of magnesium (1547.25 ± 80.44 ppm), zinc (148.62 ± 3.87 ppm), and iron (262.17 ± 10.71 ppm).

3.2 | Antioxidant capacity of BJ

The antioxidant activity of BJ was assessed through two antioxidant assays: 2-diphenyl-1-picrylhydrazyl (DPPH) radical scavenging activity and chelating capacity extraction, as detailed in Table 1. The DPPH free radical-scavenging capacities of BJ and ascorbic acid (used as a positive control) are depicted at various concentrations. The results indicate that BJ exhibits a significantly higher ability ($p < 0.05$) in a dose-dependent manner. The study also determined

TABLE 1 Phytochemical analysis, mineral composition and radical-scavenging activity of Beetroot juice (BJ).

Parameters	BJ contents
Total polyphenols (mg GAE/L)	974.21 ± 13.62
Flavonoides (mg QtE/L)	282.47 ± 5.22
Total tannins (mg TAE/L)	149.45 ± 1.57
Condensed tannin (mg TAE/L)	22.81 ± 1.98
Dietary fibers (g/L)	14.74 ± 1.25
Zinc (ppm)	148.62 ± 3.87
Copper (ppm)	22.76 ± 2.25
Manganese (ppm)	32.89 ± 3.12
Molybdenum (ppm)	1.12 ± 0.21
Iron (ppm)	262.17 ± 10.71
Magnesium (ppm)	1547.25 ± 80.44
DPPH IC_{50} (μg/mL)	90.87 ± 4.07
Metal Chelating IC_{50} (μg/mL)	120.02 ± 4.99
Ascorbic acid (μg/mL)	84.22 ± 2.98

Note: Data are expressed as mean ± SEM ($n = 3$).
Abbreviations: CGE, cyanidine glucosyl-3 equivalent; g, gramme; GAE, gallic acid equivalent; IC50, inhibitory concentration; L, Liter; QtE, quercetin equivalent; SEM, standard error of the mean; TAE, tannic acid equivalent.

the concentrations of BJ extract and ascorbic acid required to scavenge 50% of the DPPH radicals (IC_{50}). The IC_{50} value for BJ was found to be 48.56 ± 2.34 μg/mL, significantly higher than the standard used (19.36 ± 1.85 μg/mL).

The chelating capacity of the extract was measured by monitoring the inhibition of Fe (II)-ferrozine complex formation after incubation of BJ, at different concentrations, with divalent iron. The obtained results show that the inhibition concentration (IC_{50}) required to chelate 50% of Fe^{2+} is 120.02 ± 4.99 μg/mL for BJ and 66.47 ± 2.27 μg/mL for BHA.

3.3 | Acute oral toxicity of BJ

The acute toxicity study clearly showed that oral treatment with increasing doses of BJ (0.25–50 mL/kg, *b.w.*) did not change the behavior, sensory nervous system responses, gastrointestinal impacts, or vital signs of the mice during the manipulation. Furthermore, there were no apparent changes in body weight, consumption of water, or food intake. Moreover, no mortality or any toxic reactions were observed in all groups after 48 h of administration of BJ (data not shown). Therefore, the LD_{50} of BJ is considered higher than 50 mL/kg.

3.4 | Effect of LOP and BJ on food intake, water consumption and body weight

The change in food intake and body weight of the different groups between days 0 and 7 was determined, and the results are presented in Figure 1A,B. LOP caused significant ($p < 0.05$) body weight loss compared to the control group throughout the 7 days of the experiment. However, the BJ or YOH groups showed a significant increase ($p < 0.05$) in a dose-dependent manner, which followed an increase in food intake. Suggesting that the increase in body weight gain was primarily related to the enhanced nutritional intake.

The water consumption was recorded throughout the treatment period. As illustrated in Figure 1C, no significant difference ($p > 0.05$) was observed in water consumption between the various groups.

3.5 | Effect of BJ on Gastro-intestinal transit

The distance traveled by the charcoal meal during the gastrointestinal transit tests serves as an indicator of the rate of food movement through the gastrointestinal system, specifically reflecting the frequency of peristaltic contractions. The data obtained from the gastrointestinal transit (GIT) tests are presented in Figure 1D. As anticipated, a significant decrease in the intestinal charcoal transit ratio was observed in the LOP group compared to the control rats ($p < 0.05$). However, treatment with different doses of BJ

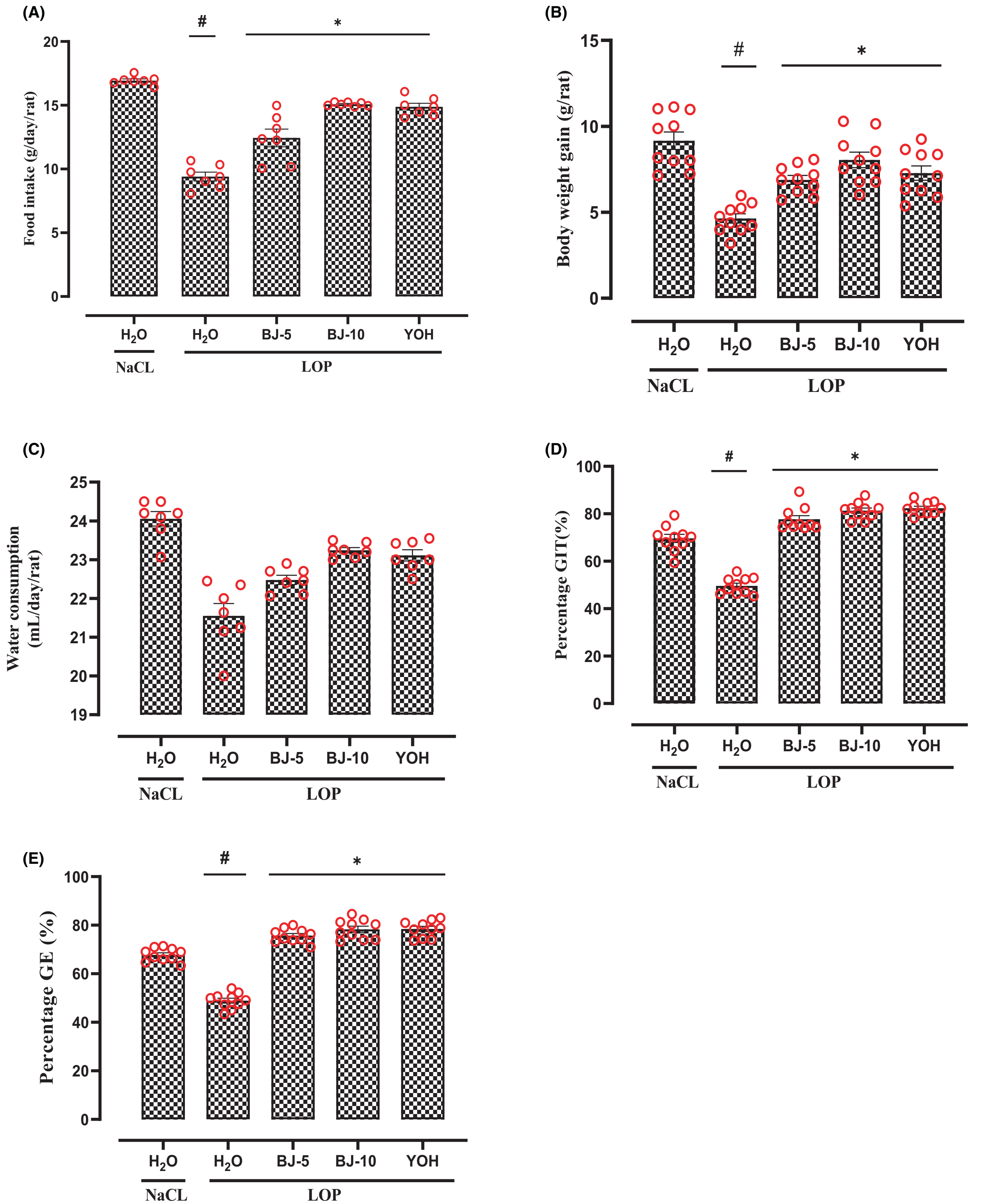

FIGURE 1 Effect of beetroot juice (BJ) and Yohimbine (YOH) on food intake (A) and body weight gain (B), water consumption (C), the acceleration of gastro-intestinal transit (D) and gastric emptying (E) during LOP-induced constipation. Animals were treated with various BJ concentrations (5 and 10 mL/kg, *b.w.*, *p.o.*), YOH (2 mg/kg, *b.w.*, *i.p.*), or NaCl (0.9%). GIT, Effect on normal intestinal transit. After 2 h of treatment, all rats received by gavage standard charcoal meal (10% charcoal in 5% gum Arabic, 1 mL/rat) and sacrificed after 30 min. GE, Effect on normal gastric emptying test meal (1.5 mL/rat) formed by 50 mg phenol red in 100 mL aqueous methyl cellulose (1.5%) was orally administrated after 1 h, and animals were sacrificed after 30 min. Data are expressed as means ± SEM (n = 10). $^{\#}p < 0.05$ versus control group and $^{*}p < 0.05$ versus LOP group (ANOVA test).

led to a significant increase ($p < 0.05$) in small intestinal transit. Moreover, these laxative effects were comparable to those of the group treated with the YOH, demonstrating a significant ($p < 0.05$) increase in the intestinal charcoal transit rate compared to the control group.

3.6 | BJ accelerates gastric emptying

Gastric emptying is a complex phenomenon that regulates the arrival of nutrients in the upper small intestine to facilitate proper digestion and absorption. As depicted in Figure 1E, a significant ($p < 0.01$) dose-dependent difference was observed between the quantity of phenol red meal emptied in the control group and that in the test groups treated with BJ or YOH. At a dosage of 10 mL/kg, BJ accelerated the emptying of the phenol red meal to 81.72%, a rate comparable to that of YOH (85.21%) and the negative control group (68.02%).

3.7 | Effect of LOP and BJ on stool parameters

As depicted in Figure 2A, a significant decrease in stool frequency was observed in the LOP-treated group. This reduction was accompanied by a decline in the wet weight of stools indicative of a decrease in stool water content (Figure 2B). However, pretreatment with BJ (5 and 10 mL/kg) led to an increase in defecation frequency compared to the LOP group. Furthermore, the administration of YOH at 2 mg/mL increased fecal frequency, with values similar to those obtained from BJ treatment. Notably, BJ treatment also provided significant protection against stool dehydration, particularly with the higher concentration of 10 mL/kg, which approached the control rate.

3.8 | Effect of BJ on lipoperoxidation

Lipid peroxidation was evaluated through MDA determination, and the outcomes are depicted in Figure 3A. MDA levels in the colonic mucosa significantly increased ($p < 0.05$) in rats treated with LOP (79.27 ± 2.34 nmol/mg protein) compared to the control group (24.83 ± 1.97 nmol/mg protein). The administration of BJ at various doses (5 and 10 mL/kg, b.w.) to the LOP-treated group significantly reduced MDA levels compared to constipated rats ($p < 0.05$).

3.9 | Effect of BJ on antioxidant enzymes activities

As depicted in Figure 3, constipation induced by LOP was linked to a substantial reduction ($p < 0.05$) in CAT (Figure 3B), SOD (Figure 3C), and GPx (Figure 3D) compared to the control group. Nevertheless, pretreatment with YOH or BJ effectively rectified the depletion of these enzymes in a dose-dependent manner. Significantly, BJ at a dose of 5 mg/kg exhibited a more pronounced effect, closely resembling the standard drug, YOH.

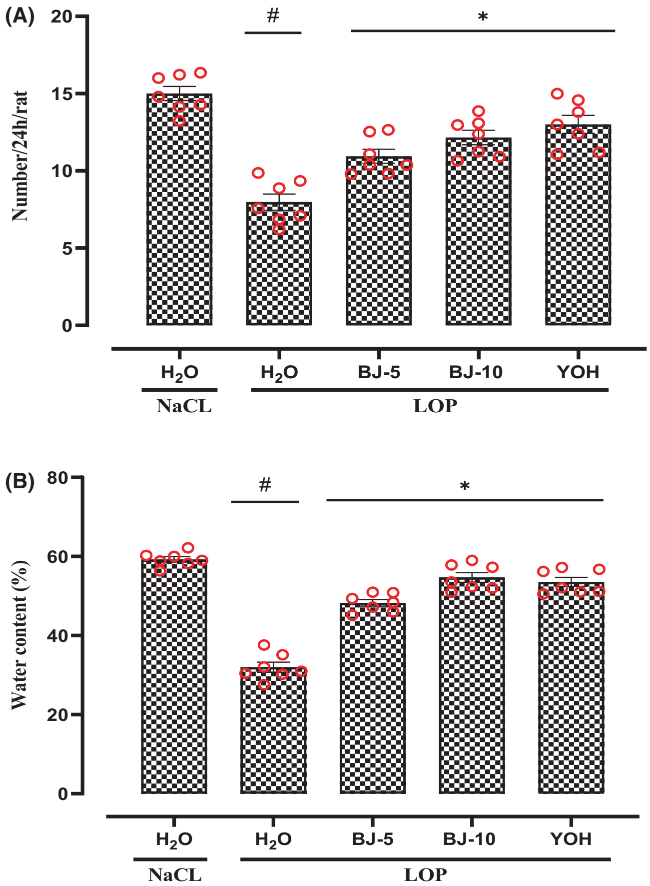

FIGURE 2 Positive influence of BJ and Yohimbine (YOH) on fecal parameters after perturbations caused by LOP administration: action on total number (A) and water fecal content (B). Animals were treated with various BJ concentrations (5 and 10 mL/kg, *b.w.*, *p.o.*), or reference molecule: YOH (2 mg/kg, *b.w.*, *i.p.*) or NaCl (0.9%) 1 h after LOP administration (3 mg/kg, *b.w.*, *p.o.*). Data are expressed as means ± SEM ($n = 10$). $^{\#}p < 0.05$ versus control group and $^{*}p < 0.05$ versus LOP group (ANOVA test).

3.10 | Effect of BJ on non-enzymatic antioxidant activities

Conversely, the impact of LOP and BJ treatments on non-enzymatic antioxidant levels in colonic mucosa was also investigated. As anticipated, LOP intoxication led to a significant decrease ($p < 0.05$) in the content of non-enzymatic antioxidants, including GSH (Figure 3E) and thiol groups (Figure 3F), when compared to the control group. Furthermore, treatments with BJ or YOH reversed the aforementioned changes, restoring them to a state close to normal. In summary, the results indicate that BJ could effectively bolster the antioxidant defense system, contributing to the mitigation of LOP-induced constipation.

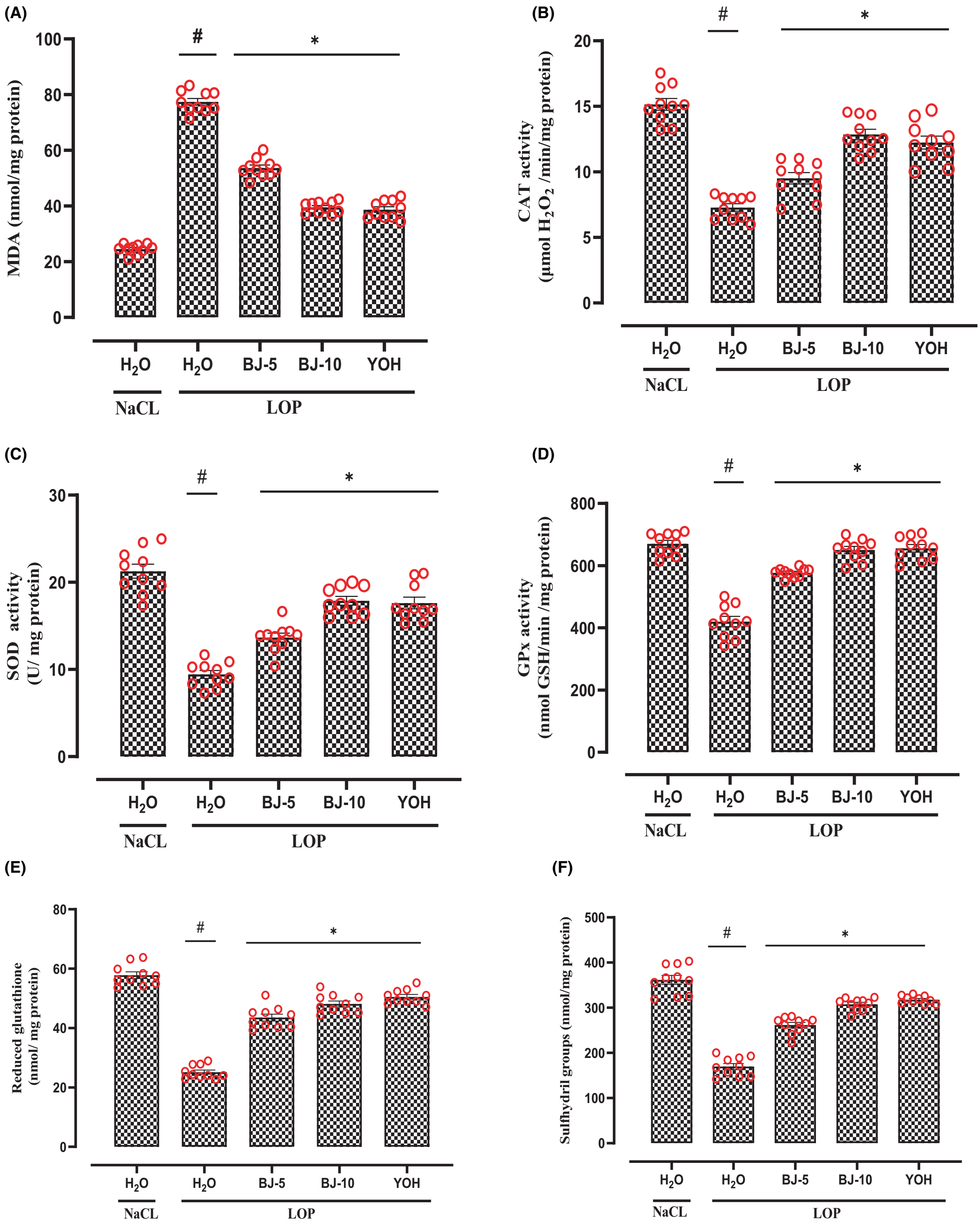

FIGURE 3 Effect of Beetroot juice (BJ) and Yohimbine (YOH) on Loperamide (LOP) induced changes on MDA (A), antioxidant enzyme activities: CAT (B), SOD (C) and GPx (D), non-enzymatic antioxidant levels: Reduced glutathione (E) and sulfhydryl groups (F) in the colonic mucosa. Animals received LOP (3 mg/kg, *b.w.*, *p.o.*), One hour after, animals treated with various BJ concentrations (5 and 10 mL/kg, *b.w.*, *p.o.*), YOH (2 mg/kg, *b.w.*, *i.p.*), or NaCl (0.9%) during a week. Data are expressed as means ± SEM ($n = 10$). # $p < 0.05$ versus control group and *$p < 0.05$ versus LOP group (ANOVA test).

3.11 | Effect of BJ on LOP-induced intracellular mediator's deregulation

The cytotoxic impact on colonic mucosal cells triggered by LOP was assessed through the disturbance of intracellular mediators such as hydrogen peroxide, free iron, and calcium, with the results presented in Table 2. Loperamide notably elevated ($p<0.05$) levels of hydrogen peroxide, free iron, and calcium in the colonic tissue. The administration of BJ significantly alleviated ($p<0.05$) all LOP-induced disturbances of intracellular mediators in a dose-dependent manner. Yohimbine, a conventional anti-diarrheal compound, is also shielded against the elevation of intracellular mediator levels induced by LOP.

3.12 | Effect of BJ on LOP-induced colonic inflammation

The impact of LOP, YOH, and various doses of BJ on the generation of pro-inflammatory cytokines such as IL-1β and TNF-α in colon tissue is illustrated in Figure 4A and total of differential leukocytes is presented in Figure 4B. As anticipated, the administration of LOP led to a significant increase ($p<0.05$) in these inflammatory parameters. However, BJ treatment provided significant protection against LOP-induced inflammation in a dose-dependent manner. Similarly, YOH restored the levels of pro-inflammatory cytokines as well as the hematological parameters to levels close to the control group.

3.13 | Effects of BJ, LOP and YOH on electrolytes responses and biochemical parameters

The impact of BJ treatment on serum electrolytes following LOP-induced constipation was examined in this study, and the results are presented in Table 3. The LOP group induced significant alterations in serum electrolytes, including sodium, potassium, and magnesium. However, treatment with various doses of BJ (5 and 10 mL/kg) or YOH demonstrated a significant improvement ($p<0.05$) in the electrolytic parameters in a dose-dependent manner.

The effects of BJ on serum lipid, liver, and kidney function biomarkers are presented in Table 3. Indeed, except for glucose level, there was no significant difference ($p>0.05$) between the different experimental groups.

3.14 | Effect of LOP and BJ on histopathological structure of distal colon

The distal colon mucosa of each group underwent histological examination to investigate associated changes resulting from the administration of LOP, BJ, and YOH. As shown in Figure 5B, LOP administration caused significant changes marked by edema, necrotic lesions, epithelial and vascular cell alterations, and inflammatory cell infiltration. However, these alterations were notably mitigated by the administration of YOH (2 mg/kg) and BJ (5 and 10 mL/kg), especially at high doses (10 mL/kg). The administration of BJ exhibited a protective effect on the colon by reducing edema, inflammation parameters, and inflammatory infiltrate (Figure 5C,D,E).

4 | DISCUSSION

In recent years, the application of natural ingredients and medicinal plants as possible therapeutic treatments for constipation and related disorders has received increased attention in pharmacological research.[45] The present investigation examined the phytochemical characteristics and laxative effects of beetroot juice (BJ) on gastric emptying, the gastrointestinal transit (GIT), and constipation induced by LOP.

Firstly, the obtained results showed that BJ exhibits a potent $DPPH^{\bullet}$ radical scavenging activity ($IC_{50}=49.72\pm3.84\,\mu L/mL$), but still lower than that of ascorbic acid ($IC_{50}=22.06\pm1.55\,\mu g/mL$). Additionally, BJ demonstrates a high chelating capacity on Fe^{2+} ions, estimated at $120.02\pm4.99\,\mu g/mL$. The antioxidant properties of BJ are likely associated with its richness in phenolic compounds. The phytochemical screening showed that BJ contains a substantial concentration of flavonols and total tannins, with a moderate concentration of total anthocyanins and carotenoids. Previous studies have indicated that these bioactive compounds

TABLE 2 Effect of beetroot juice (BJ) and Yohimbine (YOH) on LOP-induced changes in colonic intracellular mediator: Hydrogen peroxide (H_2O_2), free iron and calcium levels in rats.

Treatment	H_2O_2 (mmol/mg protein)	Calcium (mmol/mg protein)	Free iron (μmol/mg protein)
NaCl	6.18 ± 0.96	5.86 ± 0.16	9.41 ± 0.35
LOP	$21\pm1.58^{\#}$	$10.71\pm1.27^{\#}$	$19.28\pm1.75^{\#}$
BJ-5 + LOP	$15.17\pm1.03^{*}$	$9.02\pm0.54^{*}$	$14.66\pm1.92^{*}$
BJ-10 + LOP++++mL/kg	$12.86\pm1.92^{*}$	$7.36\pm0.33^{*}$	$12.29\pm0.83^{*}$
YOH + LOP	$11.91\pm1.39^{*}$	$7.43\pm0.29^{*}$	$12.77\pm0.49^{*}$

Note: Animals received LOP (3 mg/kg, *b.w.*, *p.o.*) then treated with various BJ concentrations (5 and 10 mL/kg, *b.w.*, *p.o.*), or YOH (2 mg/kg, *b.w.*, *p.o.*) or NaCl (0.9%) over 7 days. Data are expressed as means ± SEM ($n=10$). $^{\#}p<0.05$ versus control group and $^{*}p<0.05$ versus LOP group (ANOVA test).

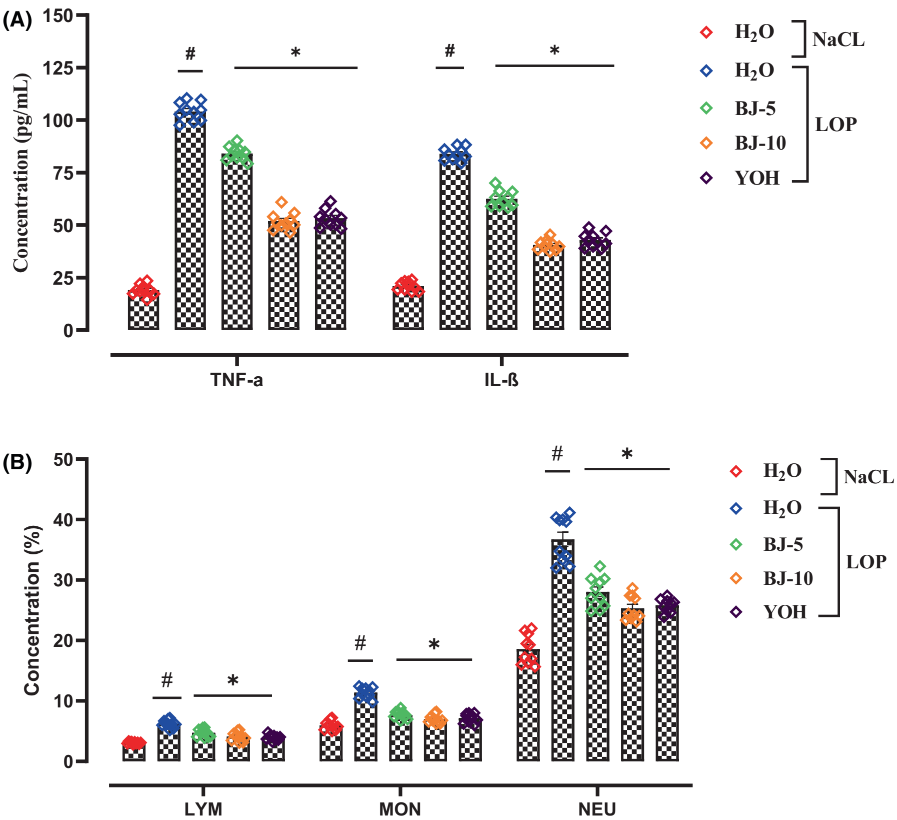

FIGURE 4 Effect of beetroot juice (BJ) and yohimbine (YOH) on colonic level of TNF-α, IL-1β (A) and Hematological Changes (B) during LOP-induced constipation (LYM, lymphocytes; MON, monocytes; NEU, neutrophiles). Animals received LOP (3 mg/kg, *b.w*, *p.o.*) One hour after, animals treated with various BJ concentrations (5 and 10 mL/kg, *b.w.*, *p.o.*), YOH (2 mg/kg, *b.w.*, *i.p.*), or NaCl (0.9%) during a week. Data are expressed as means ± SEM ($n = 10$). $^{\#}p < 0.05$ versus control group and $^{*}p < 0.05$ versus LOP group (ANOVA test).

TABLE 3 Effect of Beetroot Juice (BJ) and Yohimbine (YOH) on serum metabolic and electrolytes parameters against LOP-induced constipation in rat.

Treatment	NaCl	LOP	BJ-5 + LOP	BJ-10 + LOP	YOH + LOP
Amylase (U/L)	217 ± 16.26	222 ± 14.89	221.41 ± 10.23	216.71 ± 24.39	210 ± 12.23
Urea (mmol/L)	3.47 ± 0.23	3.99 ± 0.07	3.51 ± 0.17	3.67 ± 0.28	3.34 ± 0.25
Creatinine (μmol/L)	57.5 ± 2.27	61.25 ± 3.89	60.22 ± 2.72	59.99 ± 3.78	60.77 ± 2.22
ASAT (U/L)	85.25 ± 3.98	90.27 ± 5.55	88.11 ± 2.98	87.21 ± 3.03	84.29 ± 4.08
ALAT (U/L)	38.25 ± 2.23	41.28 ± 3.36	41.07 ± 2.11	40.19 ± 1.88	39.92 ± 3.34
TC (mmol/L)	1.49 ± 0.22	1.58 ± 0.09	1.52 ± 0.15	1.50 ± 0.34	1.53 ± 0.07
TG (mmol/L)	1.01 ± 0.03	1.12 ± 0.09	0.95 ± 0.02	0.93 ± 0.02	1.00 ± 0.09
HDL (mmol/L)	0.60 ± 0.08	0.67 ± 0.07	0.62 ± 0.03	0.59 ± 0.07	0.60 ± 0.05
LDL (mmol/L)	0.25 ± 0.04	0.23 ± 0.07	0.27 ± 0.04	0.3 ± 0.09	0.32 ± 0.06
Glu (mmol/L)	5.86 ± 0.38	11.03 ± 0.25$^{\#}$	7.03 ± 0.43*	6.29 ± 0.19*	6.88 ± 0.37*
Magnesium (mmol/L)	1.21 ± 0.11	1.77 ± 0.23$^{\#}$	1.39 ± 0.33*	1.28 ± 0.12*	1.29 ± 0.09*
Sodium (mmol/L)	142.80 ± 5.74	109.22 ± 6.03$^{\#}$	126.37 ± 3.21*	135.68 ± 4.55*	134.91 ± 2.27*
Potassium(mmol/L)	5.65 ± 0.22	10.97 ± 0.71$^{\#}$	8.03 ± 0.52*	6.87 ± 0.37*	6.40 ± 0.25*

Note: Animals received LOP (3 mg/kg, *b.w.*, *p.o.*) then treated by various BJ concentrations (5 and 10 mL/kg, b.w., p.o.), or YOH (2 mg/kg, *b.w.*, *p.o.*) or NaCl (0.9%) over 7 days. Data are expressed as means ± SEM ($n = 10$). $^{\#}p < 0.05$ versus control group and $^{*}p < 0.05$ versus LOP group (ANOVA test).

possess high antioxidant capacities against DPPH radicals.[46,47] Probably, flavonoids such as kaempferol, luteolin, and quercetin present a high antioxidant capacity and scavenger active oxygen species.[48,49]

The in vivo investigations revealed that the LD_{50} of BJ was higher than 50 mL/kg b.w., p.o. During the observation period, no instances of mortality or alterations in behavior were documented. The LOP intoxication induced a significant reduction in food consumption and did not affect water intake as compared to the control group. Conversely, BJ administration increased food intake in a dose-dependent manner. The present results corroborate several studies using LOP as a model for constipation.[9,50]

The assessment of fecal pellet number as well as its water content significantly decreased under LOP consumption. In addition to inhibiting intestinal secretion and slowing peristalsis, LOP binds to the opiate receptors on intestinal wall cells, enhances the enterocyte's ability to

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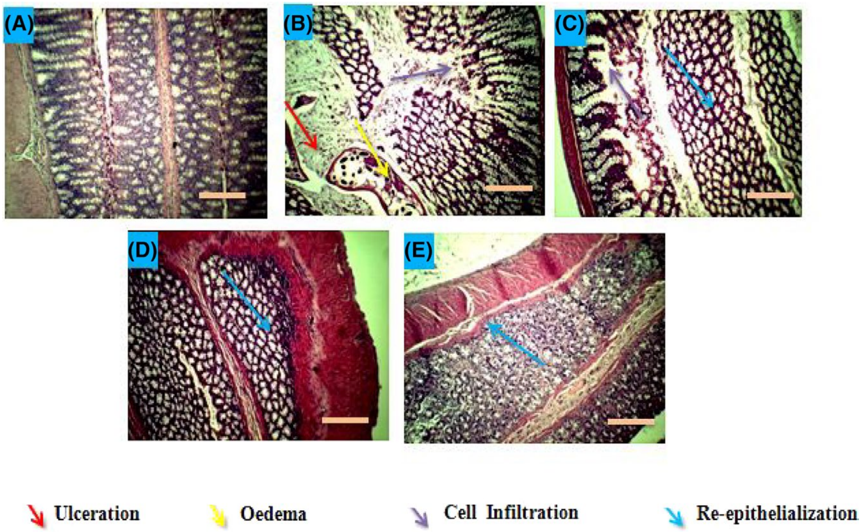

FIGURE 5 Effect of beetroot juice (BJ) and yohimbine (YOH) on colon mucosa structure and layer thickness during LOP-induced constipation in rats. Animals received LOP (3 mg/kg, b.w., p.o.) then treated with various BJ concentrations (5 and 10 mL/kg, *b.w.*, *p.o.*), Yohimbine (2 mg/kg, *b.w.*, *i.p.*) or NaCl (0.9%) during a week 1 h after LOP administration. Constipated rats (LOP) were treated only with LOP molecule (3 mg/kg, *b.w.*). Control (A), LOP (B), BJ 5 mL/kg + LOP (C), 10 mL/kg + LOP (D) and YOH + LOP (E). The histopathological modifications in the slide portions of colon tissue were analyzed by staining with H&E followed by monitoring at ×40, scale bar = 40 μm.

absorb water and electrolytes, and consequently reduces stool frequency in humans.[51,52]

Moreover, BJ administration significantly protected in a dose-dependent manner against LOP-induced hyposecretion. This protective effect was manifested through inhibiting the severity of constipation, accelerating the first onset defecation time, and stimulating wet and total stools excretion.

Several studies suggestthat BJflavonoids activate the cAMP-dependent cystic fibrosis transmembrane conductance regulator (CFTR), leading to the stimulation of chloride (Cl^-) secretion and/or the inhibition of sodium (Na^+) absorption across the colonic epithelium.[53,54]

More importantly, the administration of BJ (5 and 10 mL/kg) exerted a significant increase in gastrointestinal transit in the constipated rats. On the other hand, LOP (3 mg/kg b.w., i.p.)slowed the gastrointestinal transit and gastric emptying. Indeed, LOP reduced peristaltic frequency, leading to a decrease in hyper-secretion and volume defecation. Furthermore, LOP implicates the inhibition of acetylcholine synthesis, which results in the relaxation of intestinal muscle contractions.[55] In the same context, it has been demonstrated that this antidiarrheal compound decreases the generation of second messengers such as cyclic guanosine monophosphate (GMPc) and cyclic adenosine monophosphate (AMPc), regulates intracellular K^+ levels, and stimulates the Na^+/K^+ATPase pump.[56] However, BJ administration significantly enhanced gastric emptying, particularly with a dosage of 10 mL/kg.

Moreover, the augmentation of peristalsis, intestinal transit, and gastric emptying implies that beetroot juice influences gastric and intestinal motility by enhancing their contractile function. The stimulation of gastrointestinal motility can be attributed to the juice's rich dietary fiber content and low levels of condensed tannins. Dietary fiber is essentially comprised of carbohydrates in polymer form with three or more monomeric units, resisting digestion and absorption in the small intestine.[57] Fermentable fibers proliferate gut microbiota, leading to increased fecal biomass, which raises the colonic osmotic load, elevating the water content in feces. Various studies have documented that high-fiber foods contribute to a reduction in whole-gut transit time.[13,58] On another note, it is recognized that tannins, when combined with proteins, can form protein-tannate complexes, inducing protein denaturation. This process enhances the resilience of the intestinal mucosa and decreaseswater/electrolyteand gastrointestinal transit.[59]

Generally, intestinal motility is controlled by several factors, such as ghrelin and motilin, which accelerate gastric emptying. Tonic contractions, fundus relaxations, and antral contractions are regulated by the vagus nerve.[60]

The histopathological study revealed the structural changes in the colonic mucosa after subjecting animals to LOP treatment for a week. The observations demonstrate various alterations in the microscopic architecture of the colon. These changes were characterized by a remarkable reduction in the thickness of the mucous layer. Furthermore, LOP administration induced edema and the infiltration of inflammatory cells into the mucosa. The decrease in mucous layer thickness represents a potential risk; affecting the integrity of the mucus barrier.[61] The loss of epithelial integrity can amplify inflammatory responses.[9]

The pretreatment with BJ preserved colon epithelial integrity by maintaining mucosal thickness and goblet cells. The protective effect of BJ against structural damage LOP-induced in the colon can be explained by its phenolic compounds antioxidant properties.[62]

The present investigation, we explored how experimental constipation affects the oxidative status of the colonic mucosa. The administration of LOP (3 mg/kg *b.w.*, *i.p.*), induced oxidative stress, expressed by an increase in MDA levels, reduced enzymatic antioxidant activities (CAT, SOD, and GPx), and a decrease in non-enzymatic antioxidant contents (SH groups and GSH). BJ treatment demonstrated an important protective effect against LOP-induced oxidative stress, and the highest BJ dose showed the most preventive effect. These findings are in line with previous studies, confirming a connection between LOP-induced constipation, oxidative stress, and the laxative and antioxidant properties of plant extracts.[9,50,62]

In the same context, betanin represents the main betalainin beetroot, which is known for its antioxidant capacity.[63] Indeed, betanin free radical scavenging capacity is nearly twice that of anthocyanins.[64] Multiple studies suggest that betalains prevent cancer, cardiovascular and cerebrovascular diseases, as well as liver and kidneydamage.[65]

Besides, the in vivo antioxidant properties of BJ could also be explained by its richness in phenolic compounds. These molecules scavenged free radicals, such as hydroxyl radicals ($OH^{\bullet}$), which are the major causeof lipid peroxidation.[50] Indeed, a previous study clearly showed a positive correlation between the laxative effects and antioxidant properties of medicinal plants such as *Chrozophora tinctoria*,[66] *Dendropanax morbiferus*,[67] and *Senna macranthera*.[68]

The potential involvement of intracellular mediators in LOP-induced constipation and BJ effects was investigated. The administration of LOP led to an increase in hydrogen peroxide, free iron, and calcium levels in colonic mucosa and plasma. Constipation resulted in an accumulation of Ca^{2+} in colon cells, a phenomenon known to trigger increased ROS production, leading to a decrease in cell viability and cell death. This overload in Ca^{2+} is associated with nitric oxide synthase (NOS) activity, producing NO and increasing the risk of $ONOO^{-}$ formation.[69] Interestingly, pretreatment with BJ and YOH effectively restored Ca^{2+}homeostasis by activating the calcium pumping system. Similar mechanisms have been previously reported for other plant extracts rich in phenolic compounds, such as *Globularia alypum*.[9]

The colonic accumulation of H_2O_2 and free iron resulting from LOP catalyzes the production of toxic hydroxyl radicals ($OH^{\bullet}$) through the Fenton reaction, ultimately leading to membrane lipoperoxidation.[50]

Furthermore, the reduction in ferrous iron (Fe^{2+}) contents following BJ pretreatment can be attributed to the chelating capability of polyphenols on transition metals. This chelation process inhibits the Fenton reaction, which lowers the oxidative stress induced by the hydroxyl radical.[70,71]

The imbalance in intestinal homeostasis may be the cause of the intestinal barrierdisorders and inflammatory response. In this respect, the evaluation of the inflammatory status of the colon showed that LOP induces an inflammatory reaction characterized by increased levels of pro-inflammatory cytokines (IL-1β and TNF-α) compared to the control group. The Pretreatments with BJ and YOH exhibited a significant reduction in inflammatory marker levels. The anti-inflammatory roles of polyphenols and betanin in beet juice are potent antioxidant components, disrupting the ROS-inflammation cycle. These bioactive molecules exerted an anti-inflammatory effect by inhibiting various pathways, including cyclo-oxygenase (COX), lipoxygenase (LOX), nitric oxide synthase (iNOS), and cytokines. The potential anti-inflammatory effect of BJ phenolic compounds has been previously demonstrated in vivo and in vitro models involving the modulation of the transcription factor NF-κB.[72,73]

We have also demonstrated the positive correlation between the increase of pro-inflammatory cytokine levels and white blood cell (WBC) measurement, including lymphocytes, monocytes, neutrophils, and eosinophils. In fact, WBC plays an important role against infections and inflammatory responses.[74] An elevated WBC level represents a crucial indicator of bacterial or viral infections, fungal or parasitic infections, and inflammatory conditions such as rheumatoid arthritis, Crohn's disease, or ulcerative colitis.[75,76]

5 | CONCLUSION

In conclusion, the results clearly show that BJ provided remarkable protection against LOP-induced constipation, oxidative stress, and inflammation in rats. In particular, BJ promotes a laxative effect, facilitating colonic and gastrointestinal motilities. Additionally, our findings suggested that BJ protection can be partly attributed to its anti-inflammatory and antioxidant properties as well as its opposing effects on certain intracellular mediators. In this respect, it is interesting to use beetroot juice to treat patients with gastro-intestinal disorders.

AUTHOR CONTRIBUTIONS

A.A., N.D. and S.J. performed the phytochemical composition and the in vivo experiments. A.A., H.S, and F.A. realized the statistical analysis and conducted the in vitro and in vivo experiments. H.S. participated in data processing and experimental planning. Final draft read and approved by all authors.

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CONFLICT OF INTEREST STATEMENT

The author(s) of this article have declared that they have no potential conflicts of interest.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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